Protein Sequence Prediction

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INTRODUCTION

- This project predicts local 3D protein shapes (secondary structures) directly from 1D amino acid sequences.

- **Objective of the project:** To build a bidirectional Sequence-to-Sequence (Seq2Seq) model that classifies both granular 8-state (Q8) and functional 3-state (Q3) protein structures using evolutionary data

PROBLEM STATEMENT

- Determining a protein's 3D structure in a lab is slow and expensive, creating a massive "Sequence-Structure Gap" compared to the millions of known, easily sequenced 1D amino acid strings.

- **Why this problem is important:** Structure dictates biological function. Predicting these shapes computationally is critical for understanding diseases caused by protein misfolding (e.g., Alzheimer's) and accelerating targeted drug discovery.

LITERATURE REVIEW

- Early attempts relied on isolated single sequences and simple probability counting, which hit an accuracy ceiling of around 50–60%.
- **Key findings:** The PSIPRED model (Jones, 1999) proved that using Position-Specific Scoring Matrices (PSSMs) which act as evolutionary fingerprints from homologous proteins boosts prediction accuracy to ~80%.
- **Limitations in existing solutions:** Older models often read sequences purely left-to-right (missing critical surrounding physical forces) or failed to map complex Q8 states efficiently.

WORK PLAN

Week	Work	Status
Week 1	Requirement analysis, dataset description, tools & technology identification, and study of available research.	Completed
Week 2	Studied protein sequences, reviewed ML approaches, identified datasets, and planned workflow.	Completed
Week 3	Developed initial block diagram of the proposed system, finalized data processing, model training, and prediction flow.	Completed
Week 4	Defined and implemented evaluation metrics on test data, and compared model performance.	Completed
Week 5	Evaluated the model on unseen sequences, analyzed prediction errors, and identified areas for accuracy improvement.	Completed
Week 6	Optimize Bidirectional recurrent models (BiLSTM/BiGRU) and perform comprehensive prediction comparison.	Pending

INDIVIDUAL CONTRIBUTION

Member	Enrollment No.	Contribution
Harsh	231B131	Designed the initial backend architecture and UI flow for sequence inference, and planned the model-to-frontend integration pipeline.
Himanshu	231B141	Finalized dataset selection (CullPDB/CB513), designed the data preprocessing pipeline.
Rudransh Srivastava	231B274	Researched Q8/Q3 protein biology, conducted requirement analysis, planned the workflow, and defined evaluation metrics.

DATASET DETAILS

Property	CullPDB (Training Data)	CB513 (Testing Data)
Source	Protein Data Bank (Filtered via PISCES)	Cuff & Barton Benchmark
Total Sequences	~5,534 (Filtered Subset)	513 Proteins
Format	Numpy Arrays (`.npy` or `.h5`)	Numpy Arrays (`.npy` or `.h5`)
Input Features	Sequence (One-Hot) + PSSM Profiles	Sequence (One-Hot) + PSSM Profiles
Feature Dimensions	42 dims (21 AA + 21 PSSM scores)	42 dims (21 AA + 21 PSSM scores)
Target Classes (Q8)	8 (H, G, I, E, B, T, S, C)	8 (H, G, I, E, B, T, S, C)
Target Classes (Q3)	3 (Helix, Strand, Coil)	3 (Helix, Strand, Coil)
Most Common Class	Coil (C) / Alpha-Helix (H)	Coil (C) / Alpha-Helix (H)
Redundancy Cutoff	**< 25%** sequence identity	Independent standard test set
Sequence Length	Padded to 700 residues / sequence	Padded to 700 residues / sequence
Missing Data Handling	Masked using `NoSeq` padding	Masked using `NoSeq` padding

WORK DONE SO FAR

- **Dataset Selection & Feature Mapping:** Finalized standard benchmark datasets (CullPDB for training, CB513 for evaluation); defined the 42-dimensional input representation combining 21 one-hot amino acid tokens with 21 PSSM evolutionary conservation scores.

- **Preprocessing & Pipeline Setup:** Established strict token-level sequence padding (standardized to 700 residues), handled non-standard amino acid masking, and mapped target sequences to Q8 (8-state) and Q3 (3-state) label matrices.

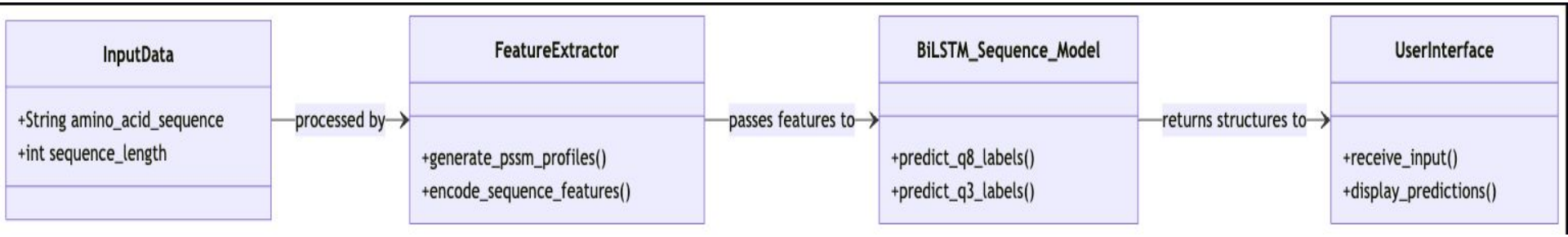
- **System Architecture Design:** Finalized the Bidirectional Recurrent architecture (BiLSTM / BiGRU) to ensure hidden states at position i capture chemical and structural context from both upstream (N-terminal) and downstream (C-terminal) neighbors.

WORK DONE SO FAR

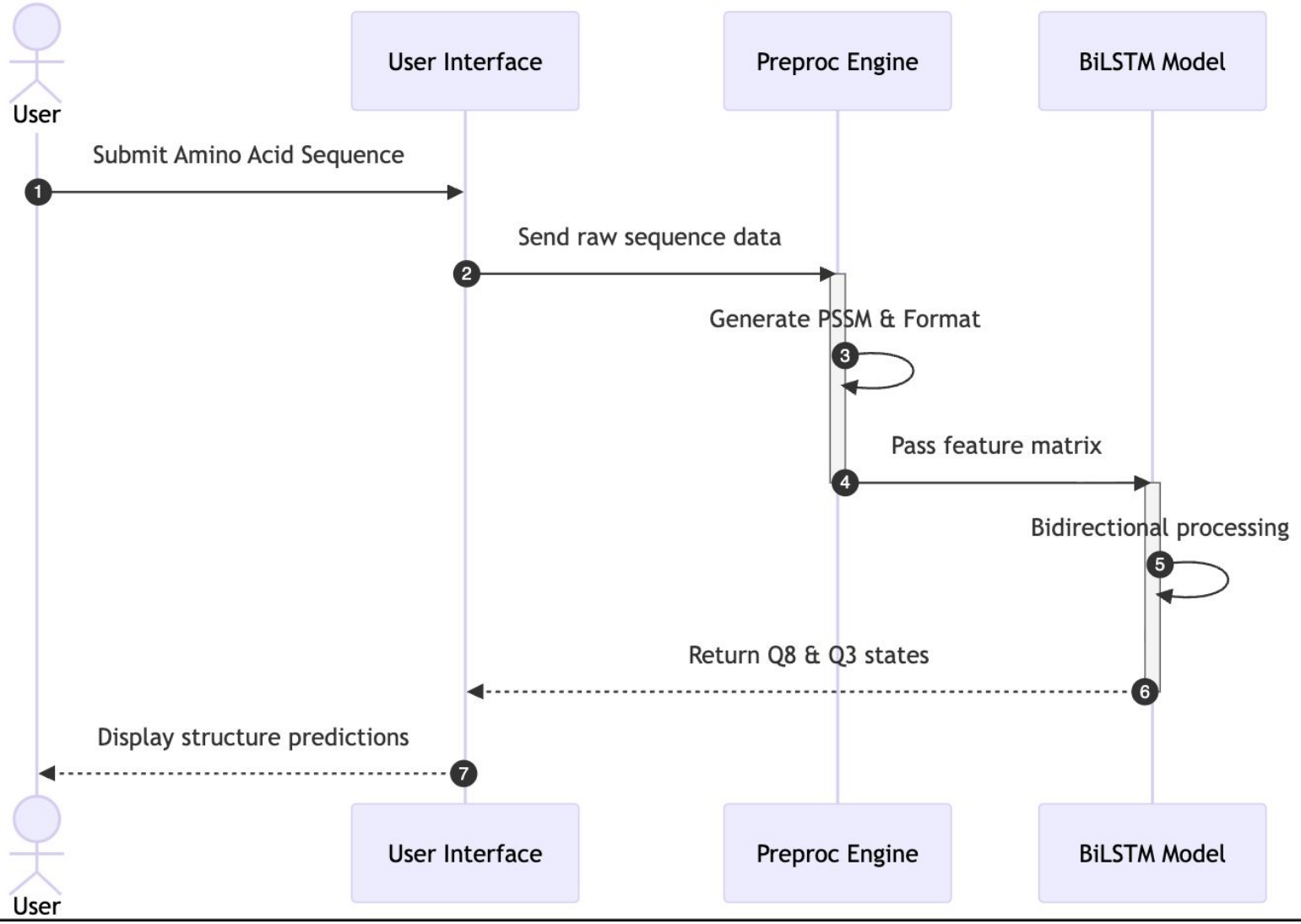
- **Evaluation Framework Ready:** Implemented and tested automated evaluation scripts tracking Q8 and Q3 accuracy, per-class precision, recall, and confusion matrices on dummy sequence tensors.

- **Application Prototype Drafted:** Outlined the backend API service (FastAPI) and front-end interface wireframe to accept raw FASTA sequences and output predicted secondary structure strings.

STRUCTURAL DIAGRAM



SEQUENCE DIAGRAM



WORK TO BE DONE

- **Model Implementation & Training:** Code the BiLSTM/BiGRU architecture, process the CullPDB dataset, and begin the actual model training and hyperparameter tuning.

- **Application Development:** Build out the FastAPI/Flask backend and frontend UI to allow users to input raw amino acid sequences (FASTA format).

- **Integration & Evaluation:** Connect the trained predictive model to the user interface and evaluate its final inference accuracy against the CB513 test dataset.

CONCLUSION

- **Summary of progress so far:** The project's foundational phase is complete. We have successfully conducted our requirement analysis, finalized standard PSSP datasets, designed the bidirectional Seq2Seq architecture, and readied our evaluation metrics.

- **Expected outcomes:** In the upcoming weeks, we will transition from design to implementation, resulting in a fully trained BiLSTM/BiGRU model and an interactive web interface capable of predicting Q8 and Q3 secondary structures directly from amino acid sequences.

REFERENCES

[[1] D. T. Jones, "Protein secondary structure prediction based on position-specific scoring matrices," *Journal of Molecular Biology*, vol. 292, no. 2, pp. 195-202, 1999. [Online]. Available: https://doi.org/10.1006/jmbi.1999.3091

[2] J. A. Cuff and G. J. Barton, "Evaluation and improvement of multiple sequence methods for protein secondary structure prediction," *Proteins: Structure, Function, and Bioinformatics*, vol. 34, no. 4, pp. 508-519, 1999. [Online]. Available: https://doi.org/10.1002/(sici)1097-0134(19990301)34:4

[3] G. Wang and R. L. Dunbrack, "PISCES: a protein sequence culling server," *Bioinformatics*, vol. 19, no. 12, pp. 1589-1591, 2003. [Online]. Available: https://doi.org/10.1093/bioinformatics/btg224

[4] S. K. Sønderby and S. W. Sønderby, "Protein secondary structure prediction with long short term memory networks," *arXiv preprint arXiv:1412.7828*, 2014. [Online]. Available: https://arxiv.org/abs/1412.7828

Thank You